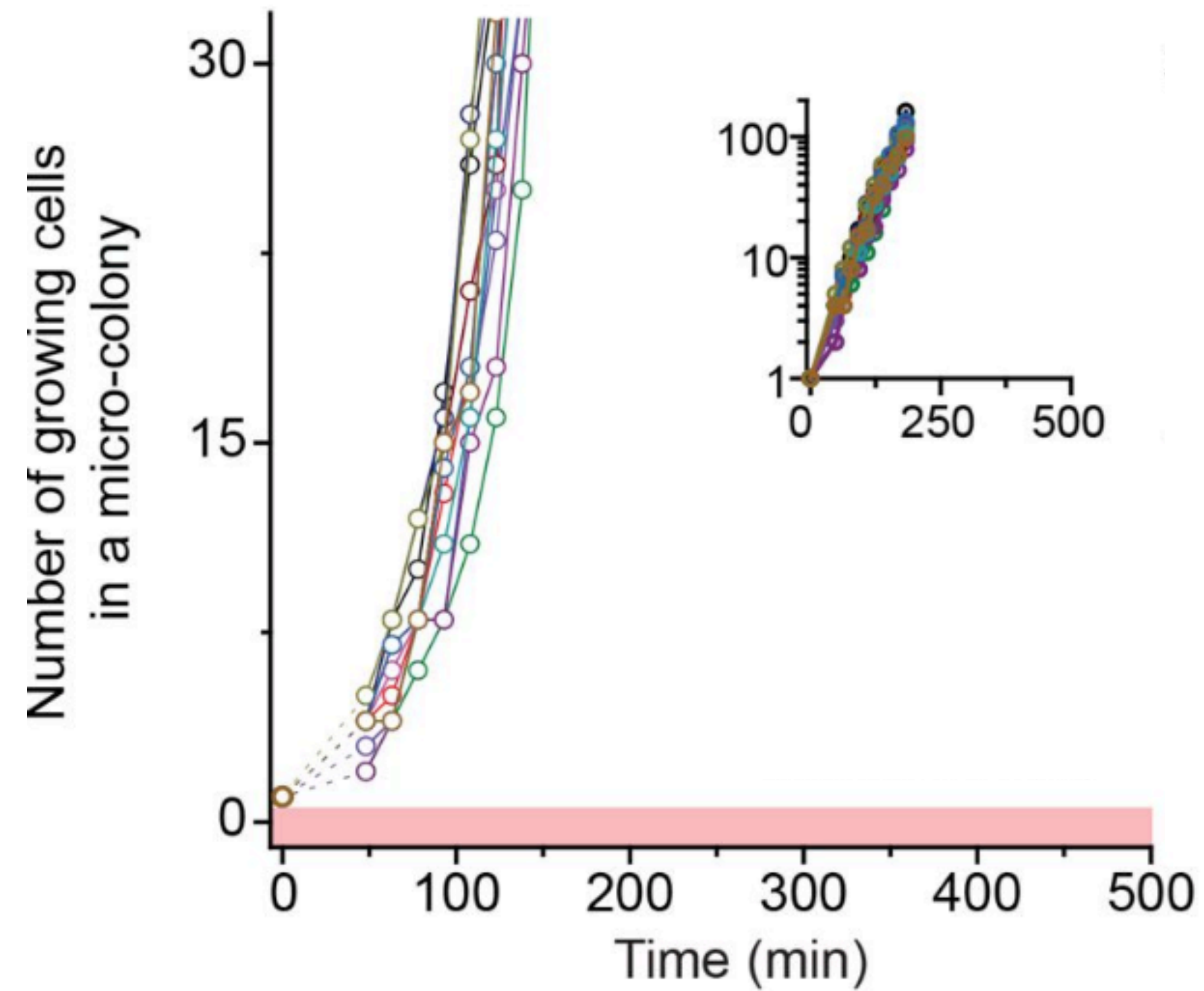
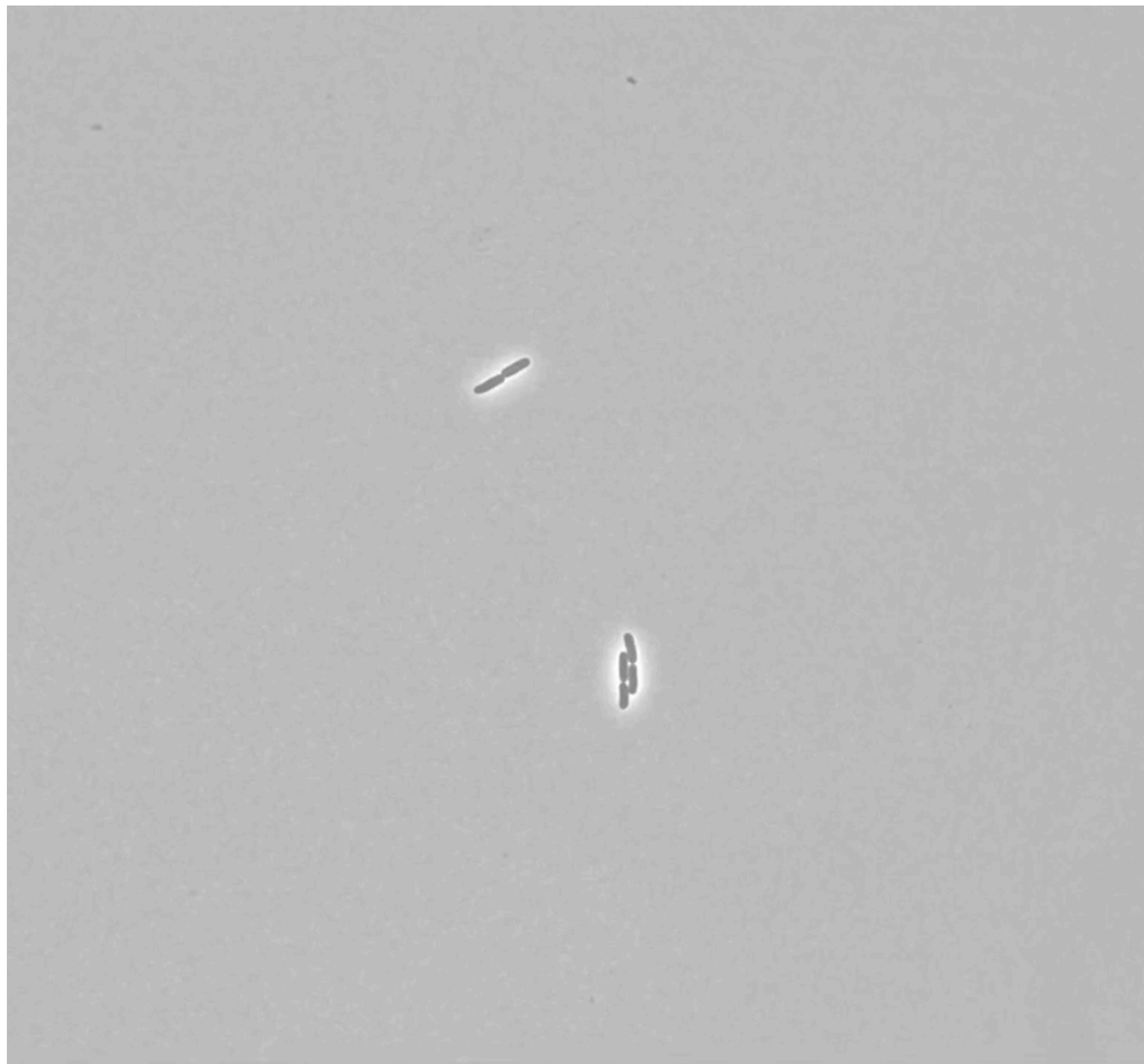


L2

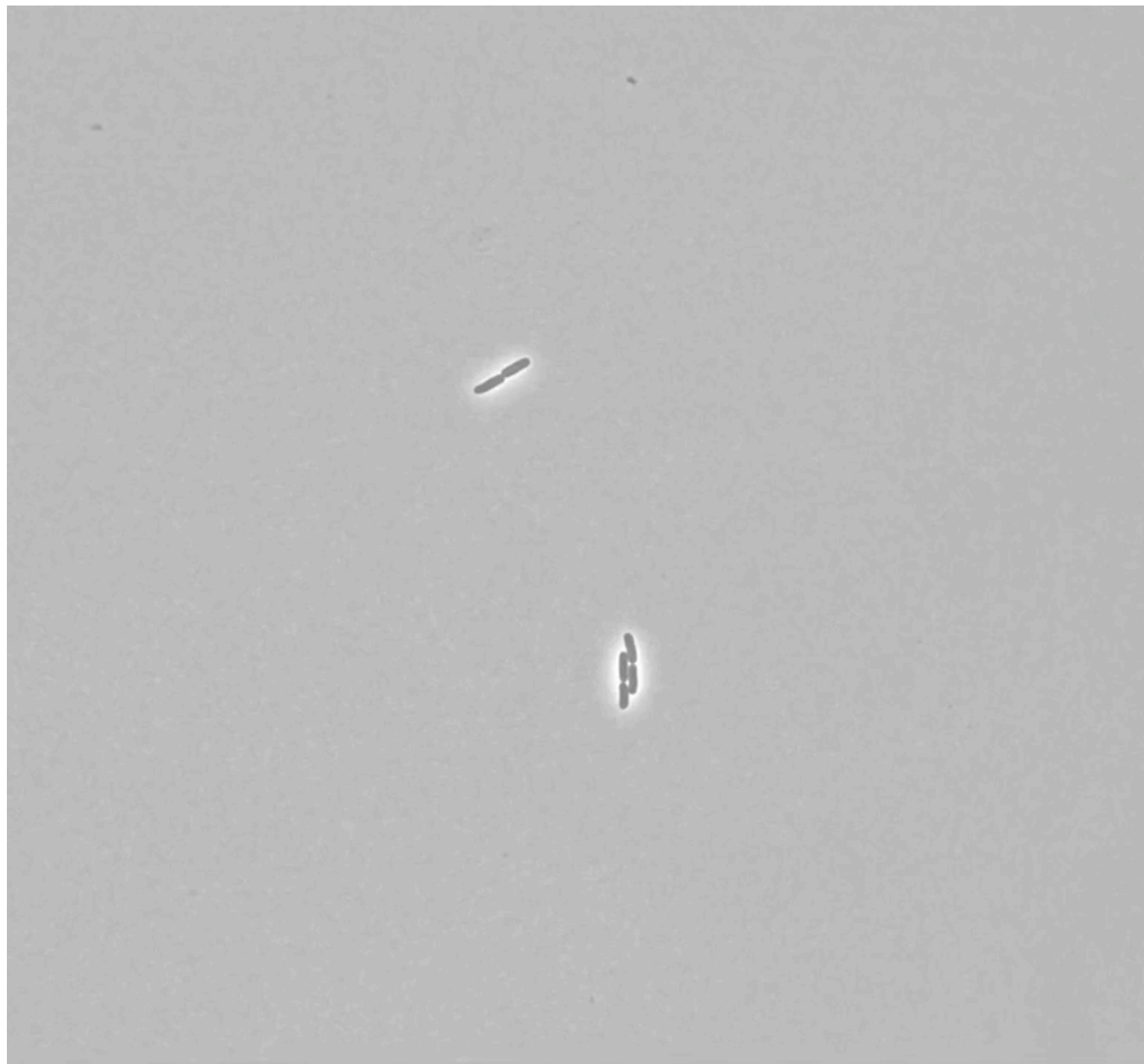
Demographic noise and stochastic extinctions

Jacopo Grilli - jgrilli@ictp.it - <http://jacopogrilli.github.io>

Modulate growth rates



Modulate growth rates



+ add antibiotics

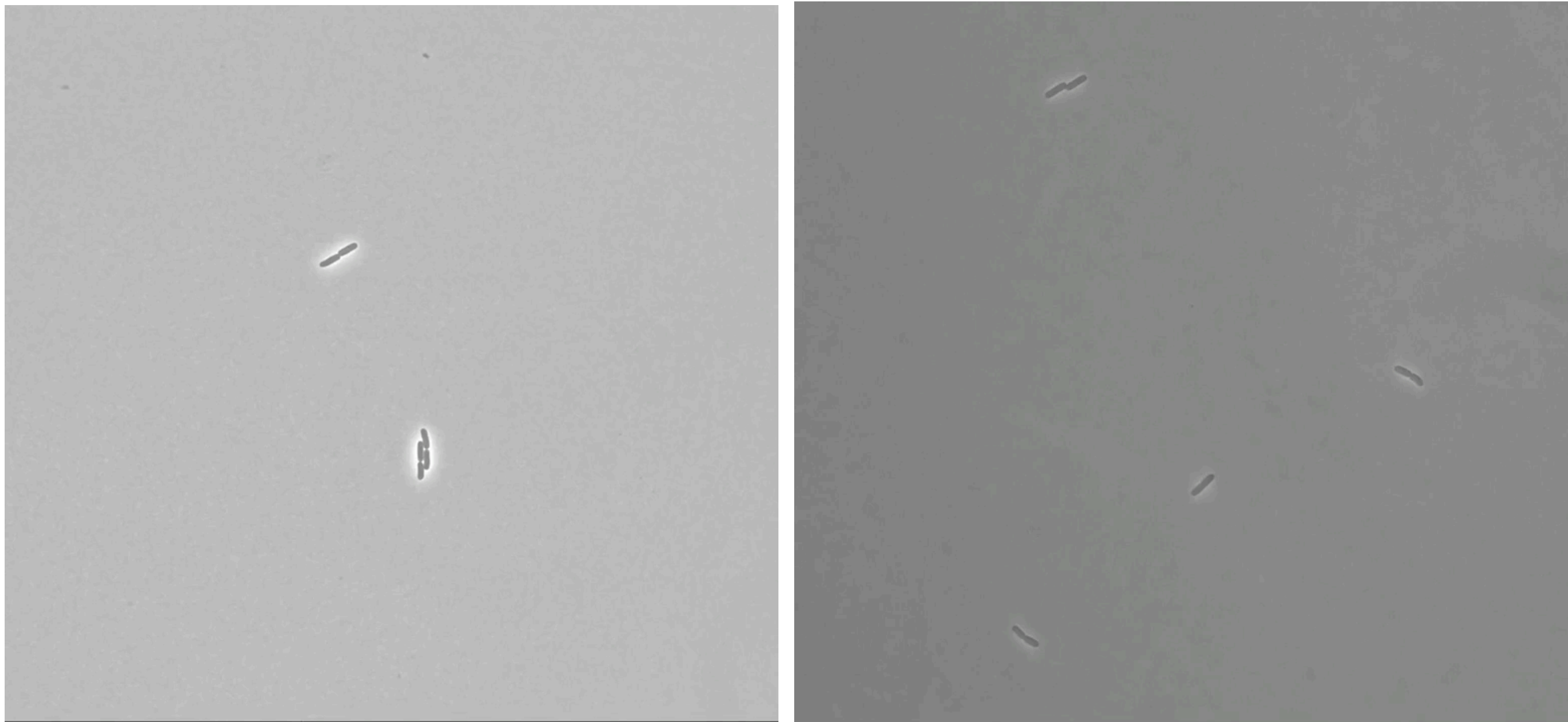
two types:

slow division (bacteriostatic)

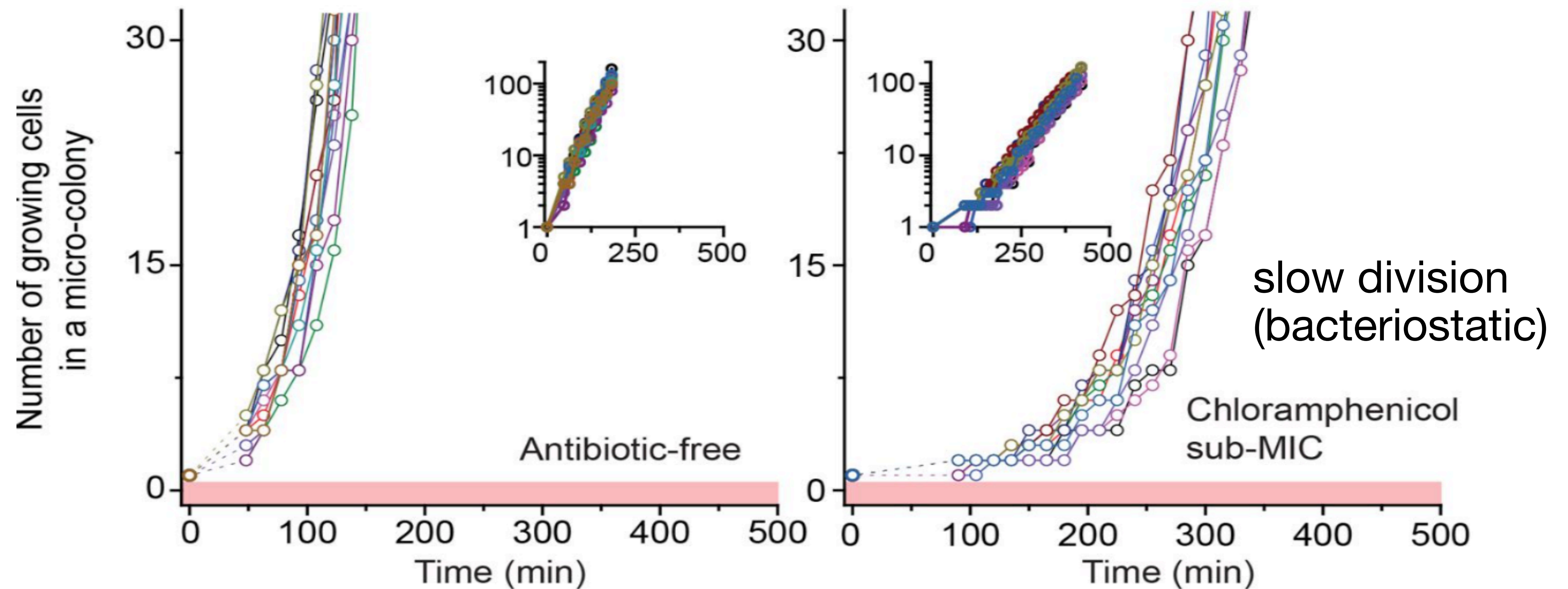
kill cells (bactericidal)

Modulate growth rates by slowing division

slow division (bacteriostatic)

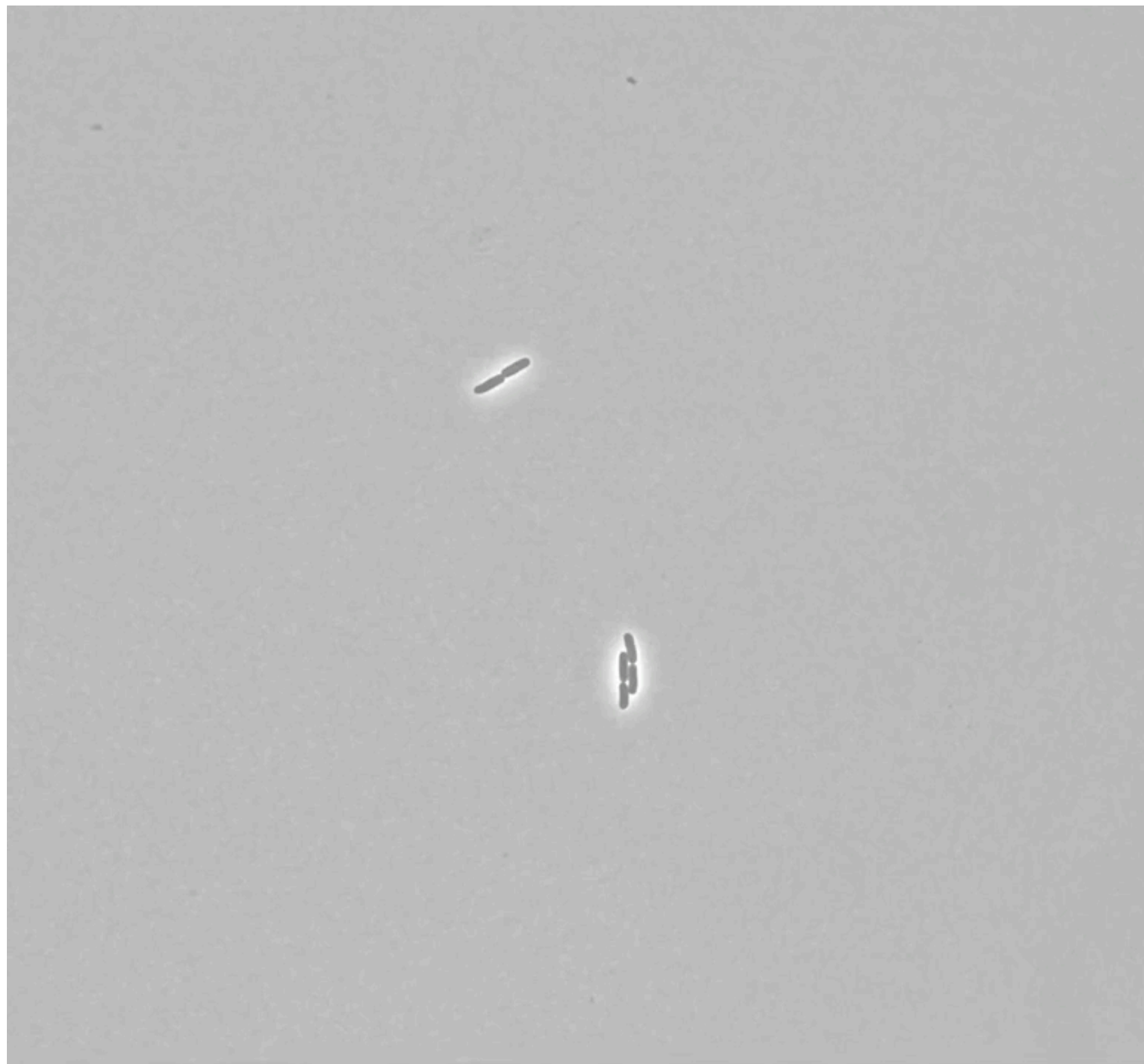


Modulate growth rates by slowing division

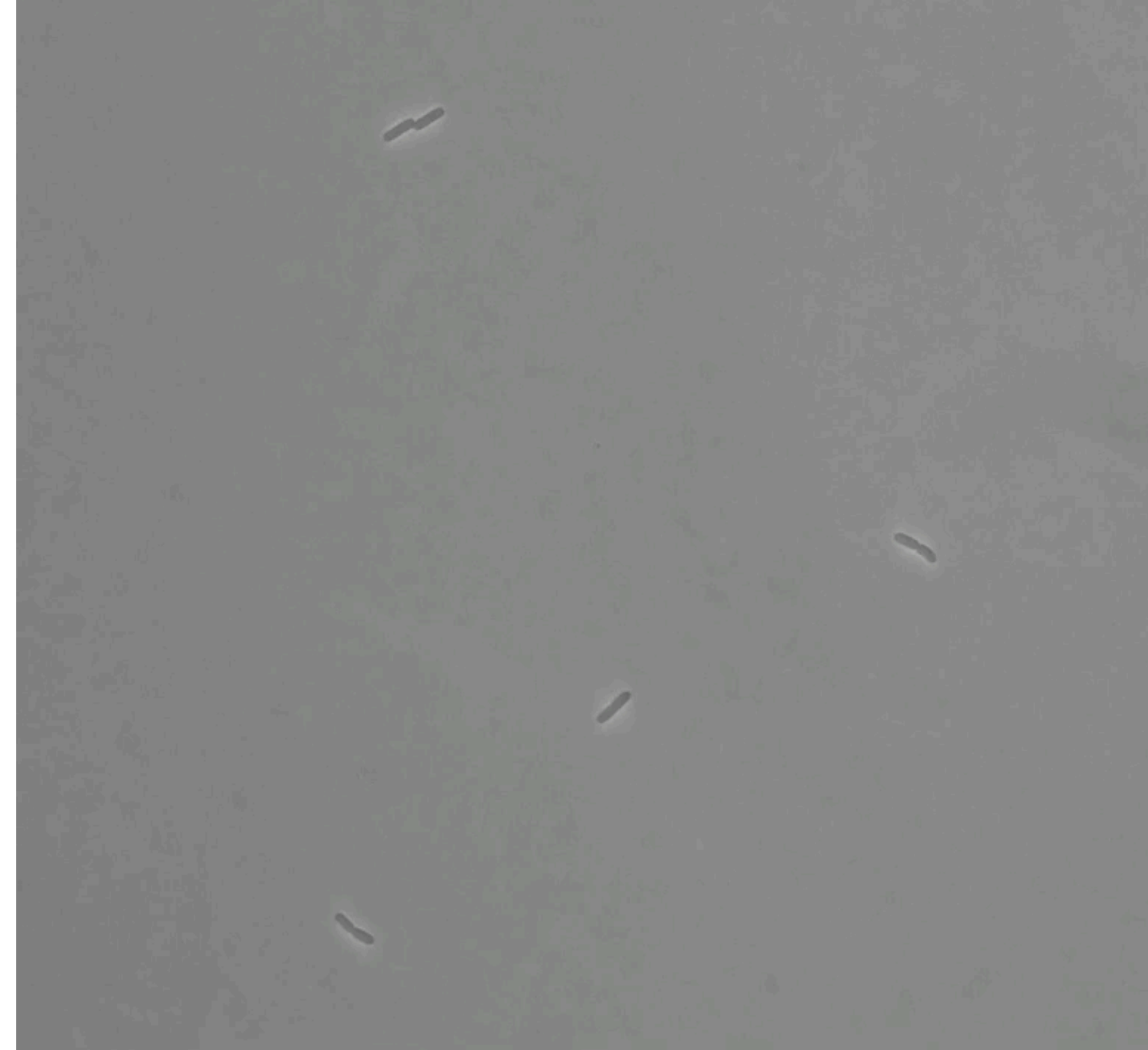


Modulate growth rates by killing cells

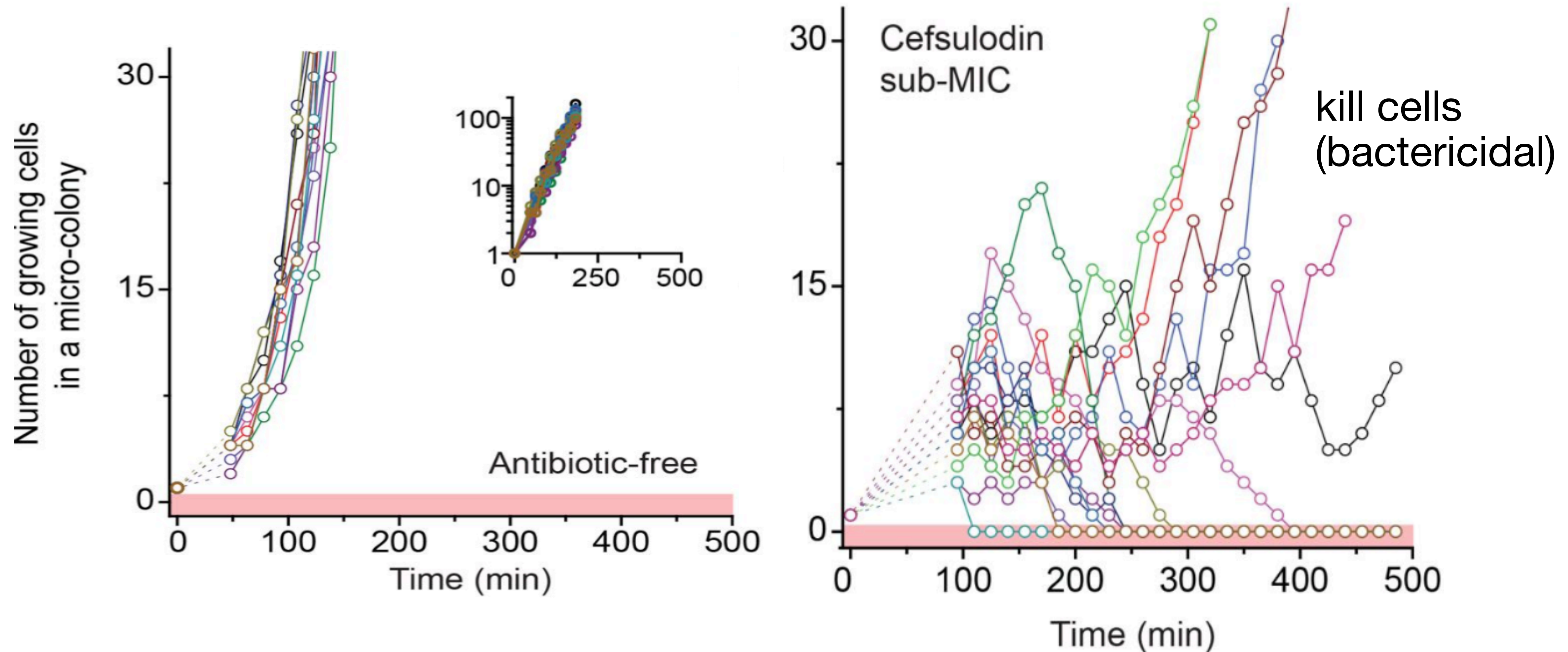
slow growth (bacteriostatic)

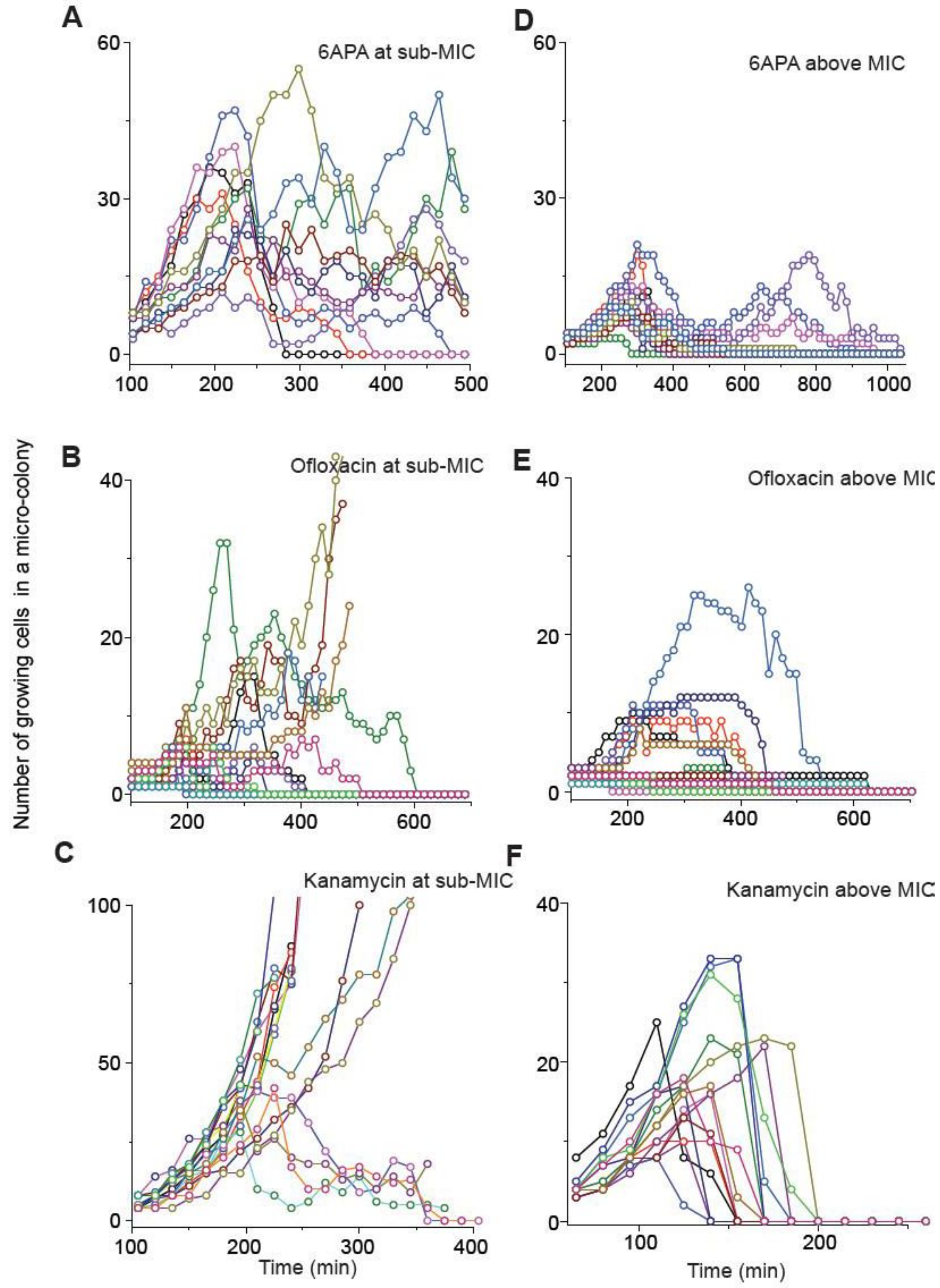


kill cells (bactericidal)

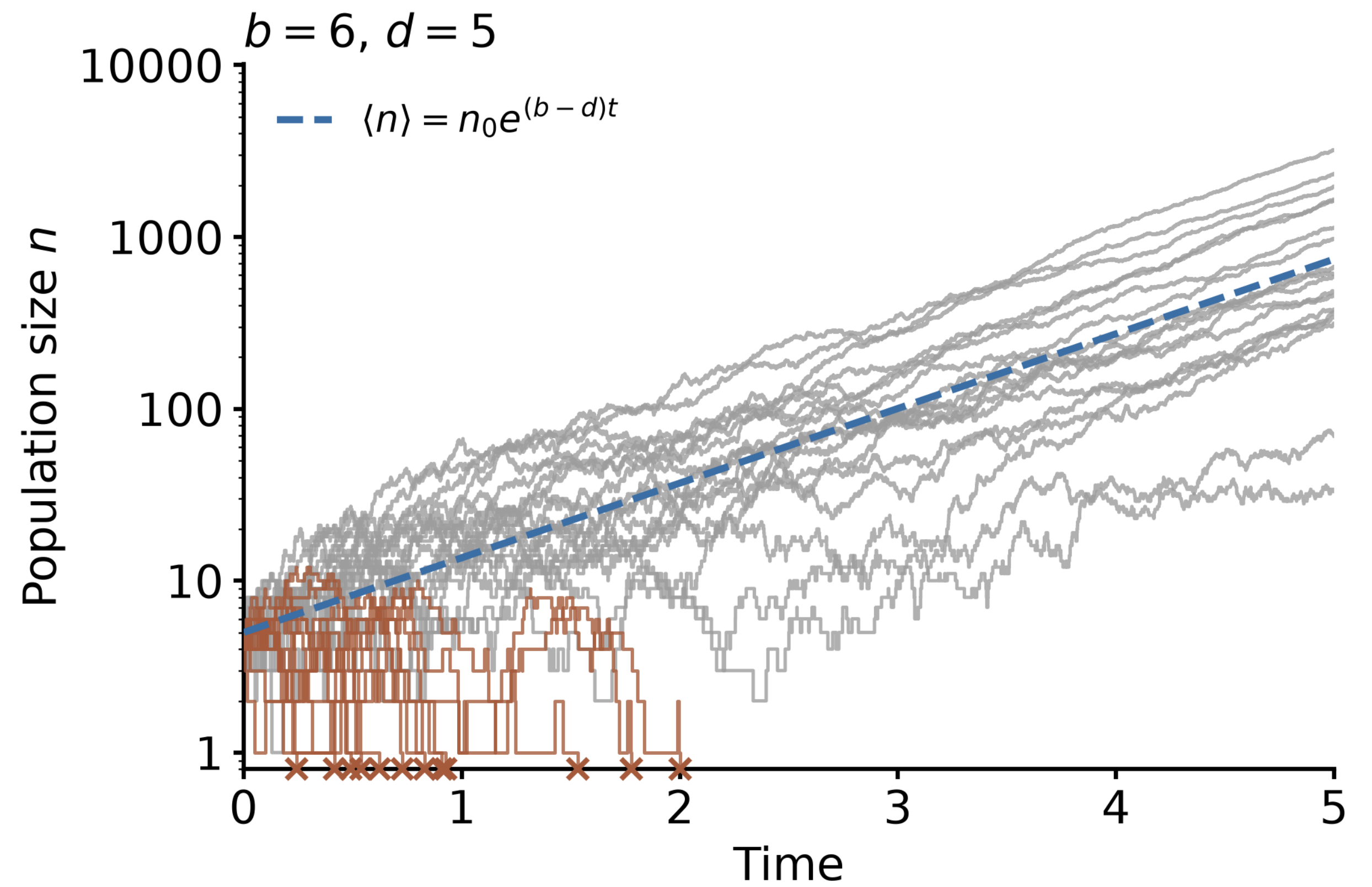
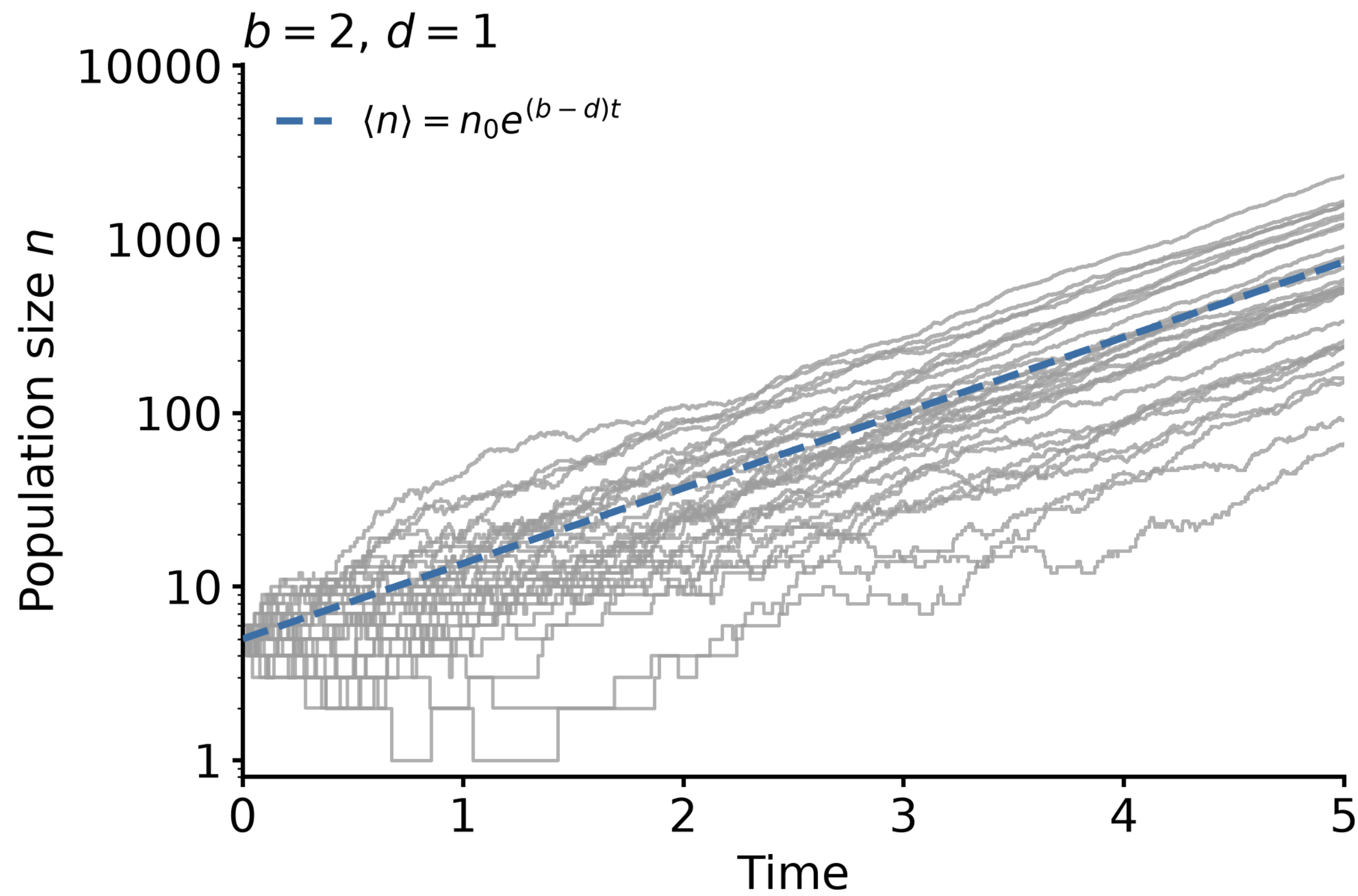


Modulate growth rates by killing cells

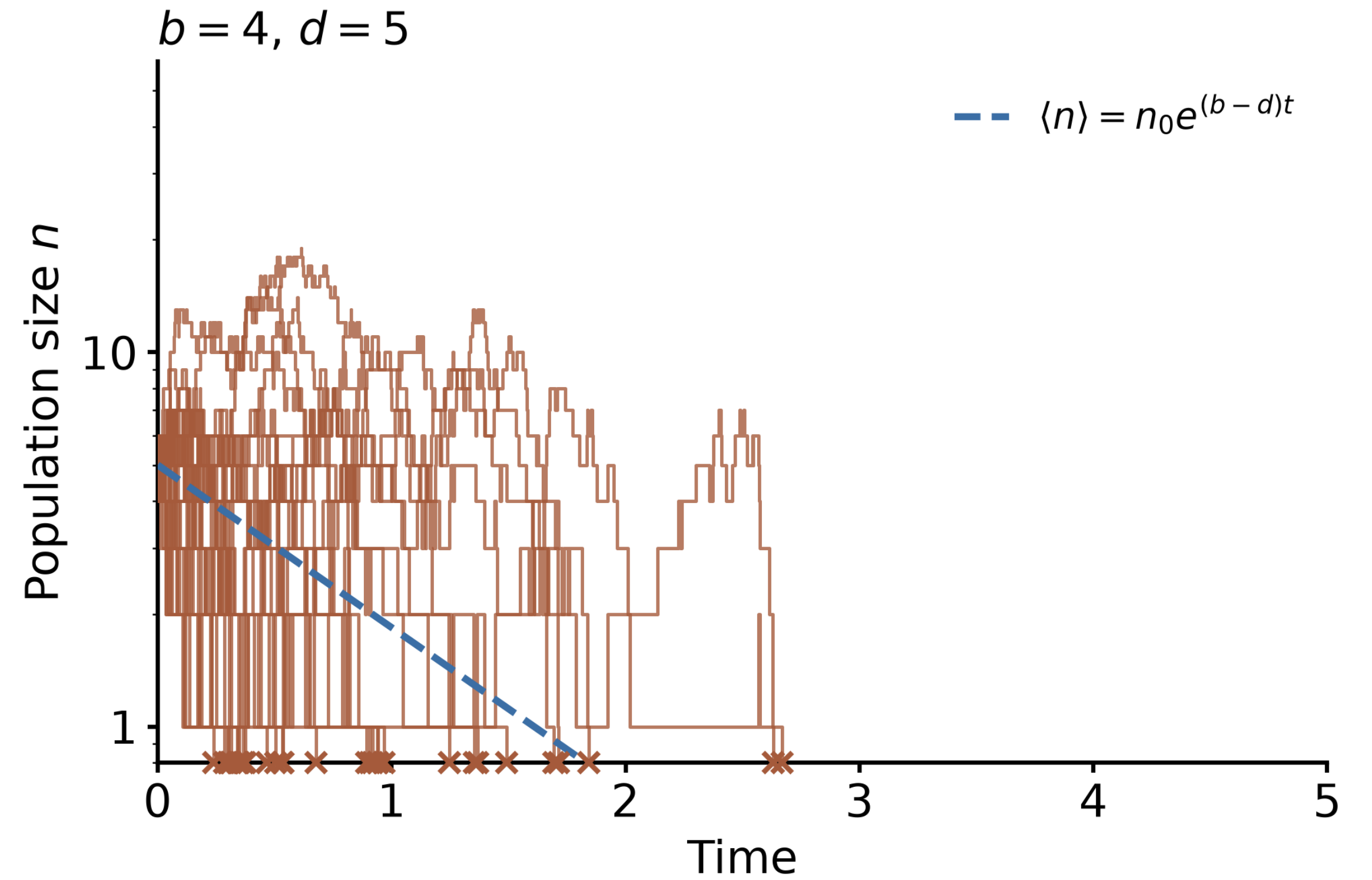
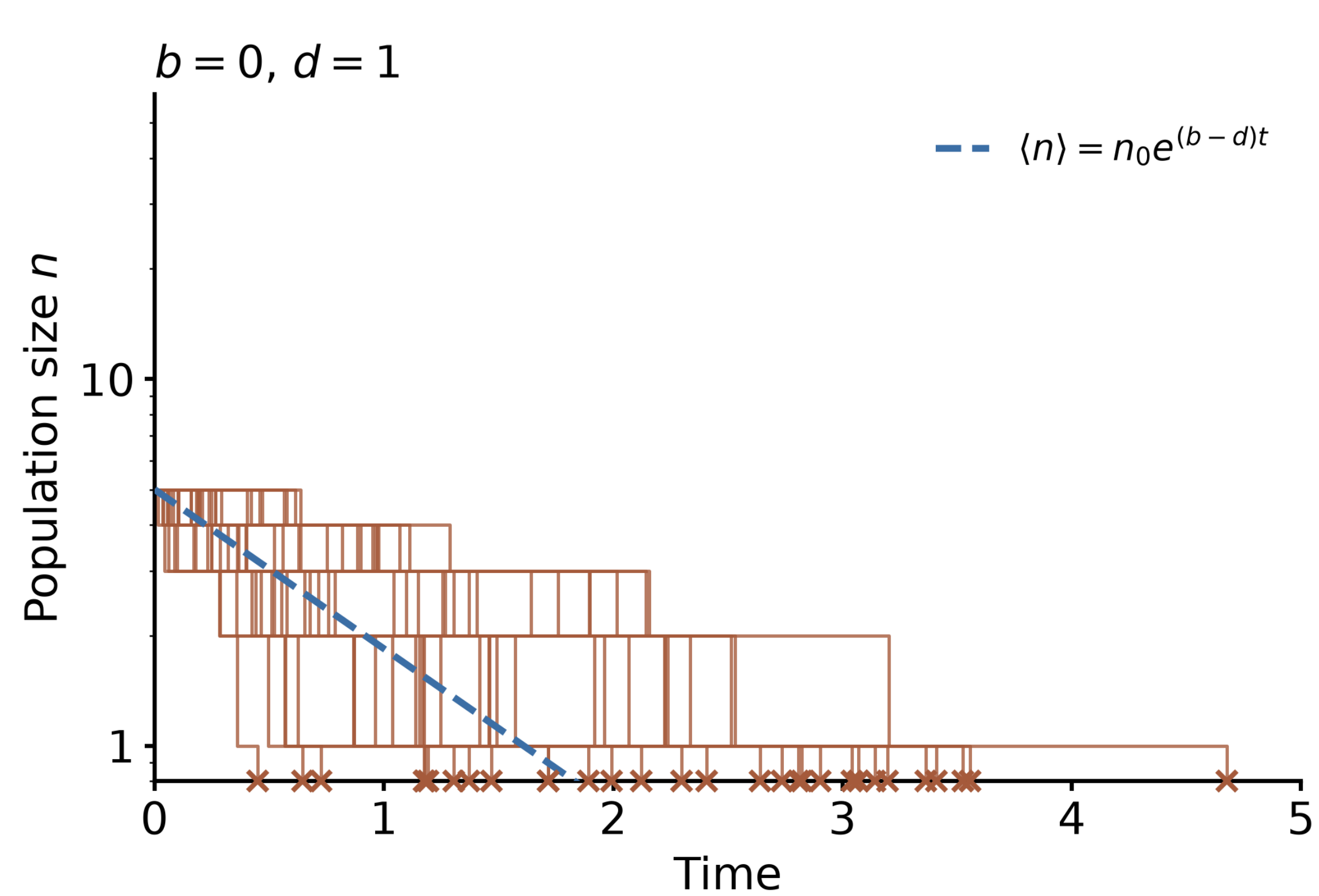




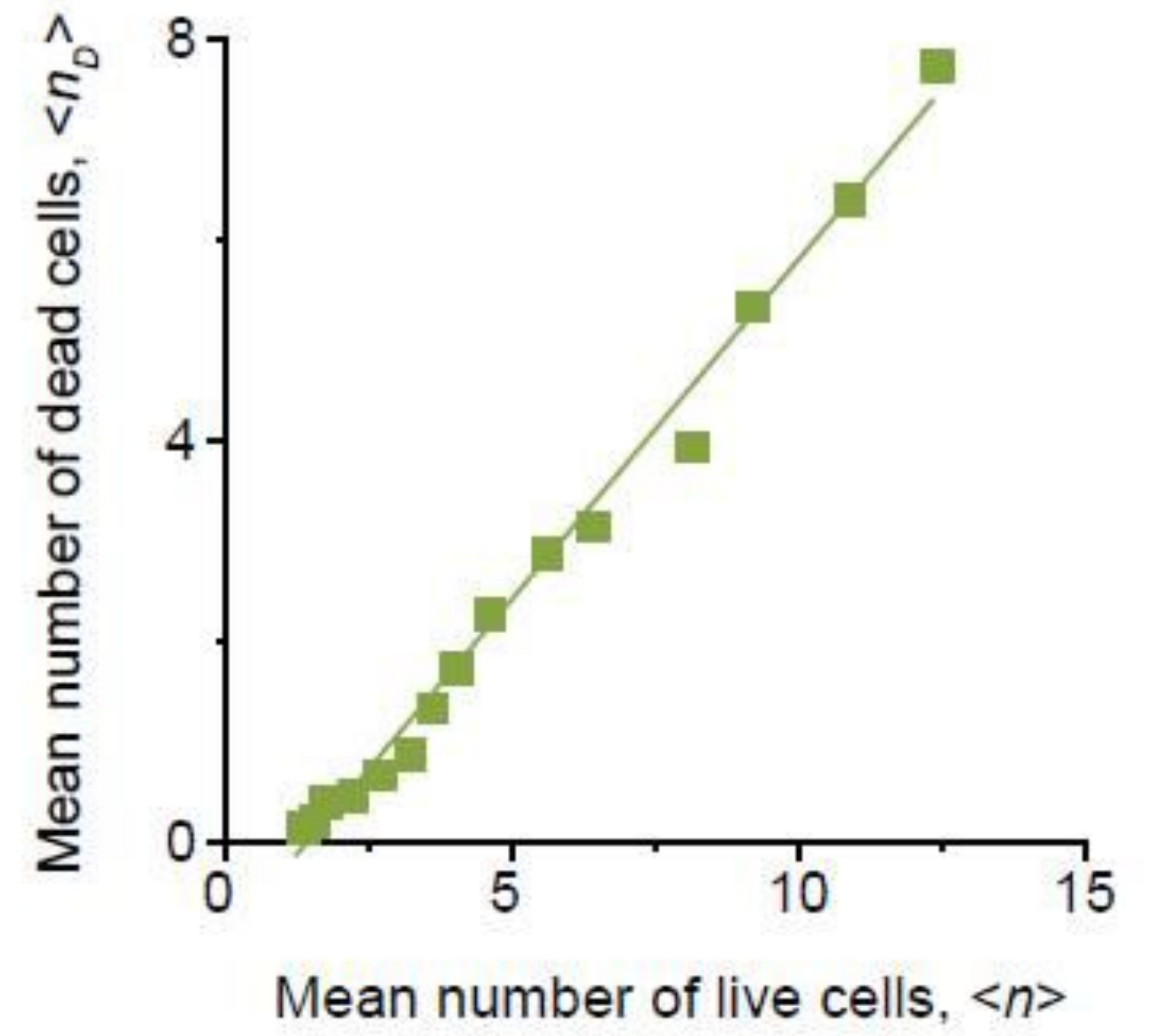
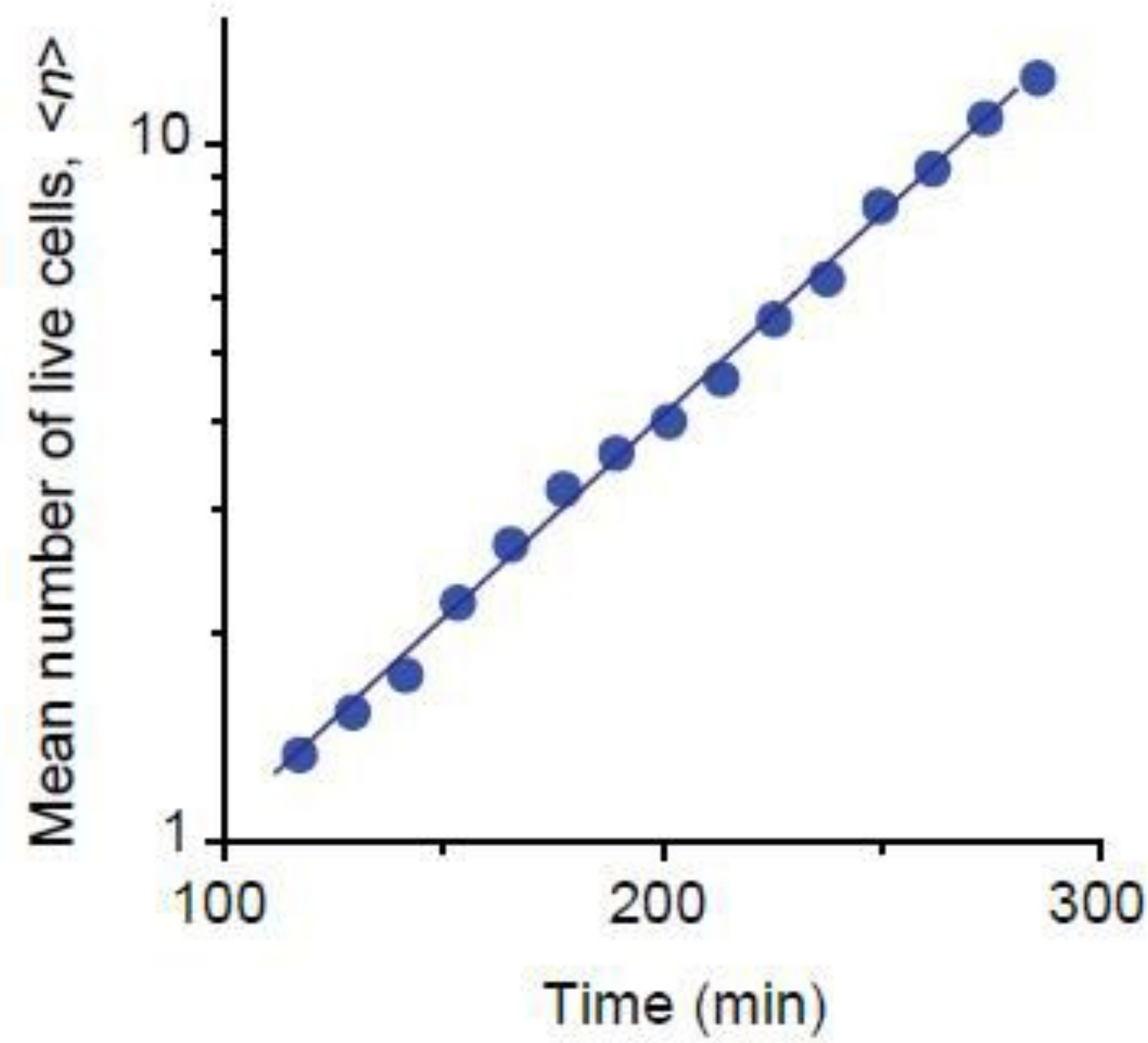
Trajectories $r=1$ different d



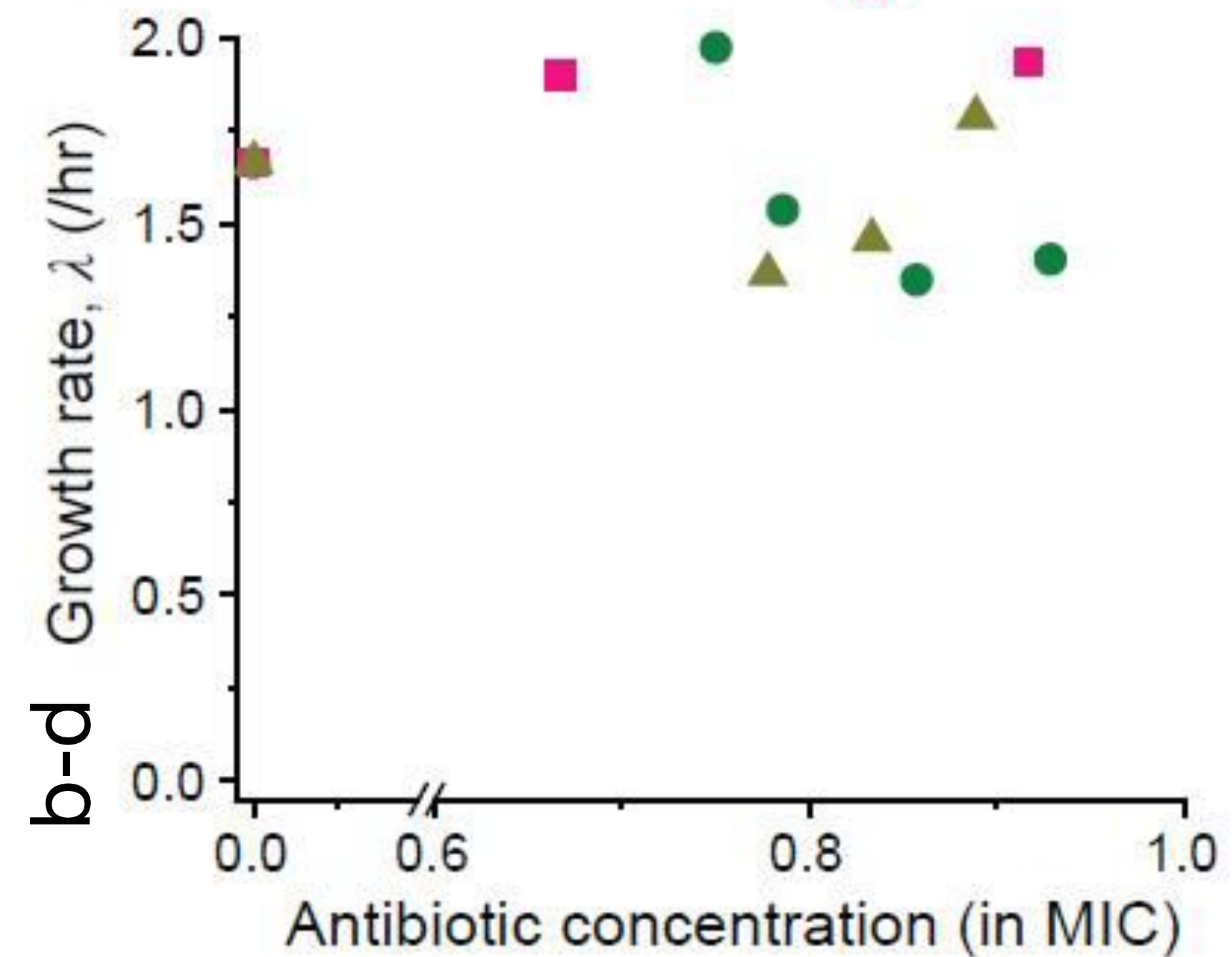
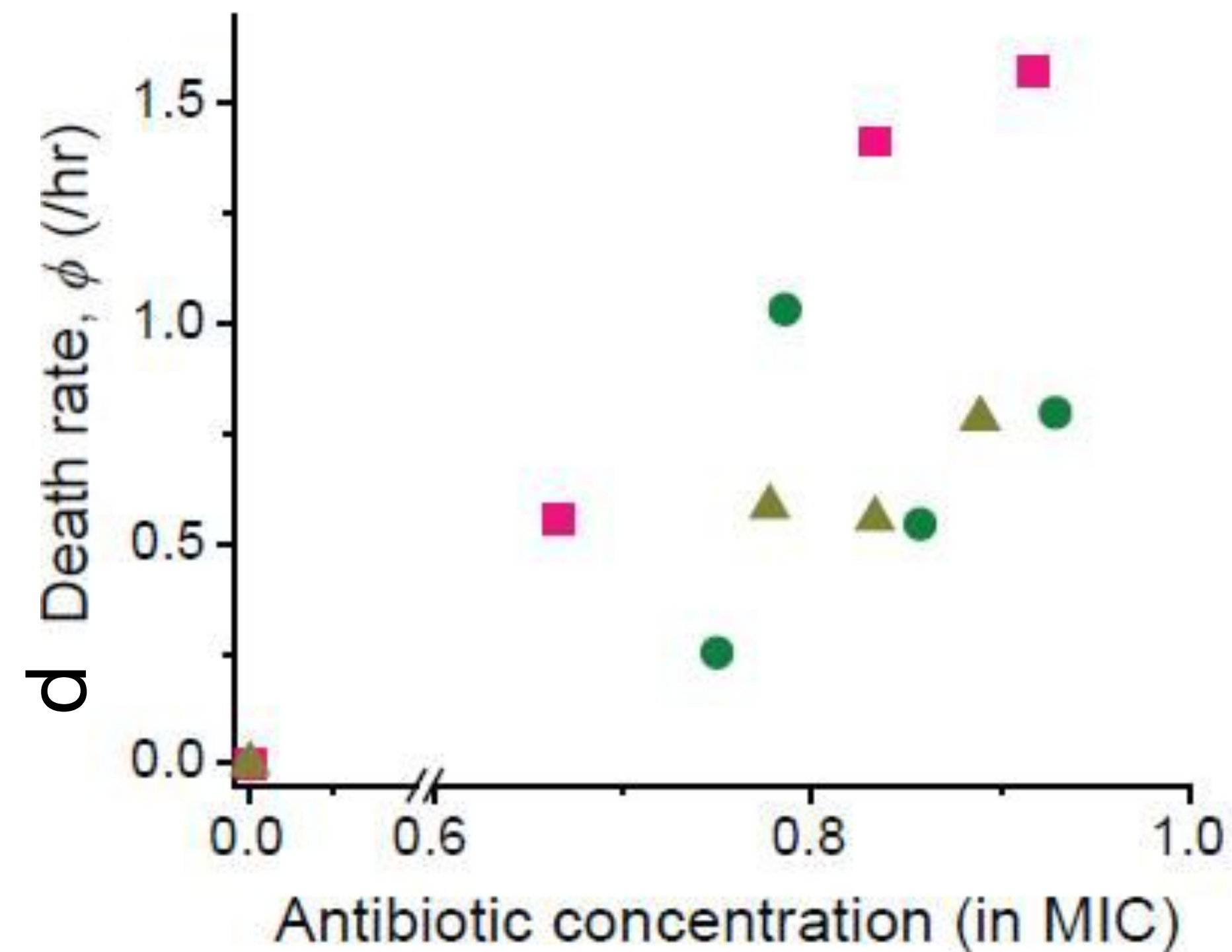
Trajectories $r=-1$ different d



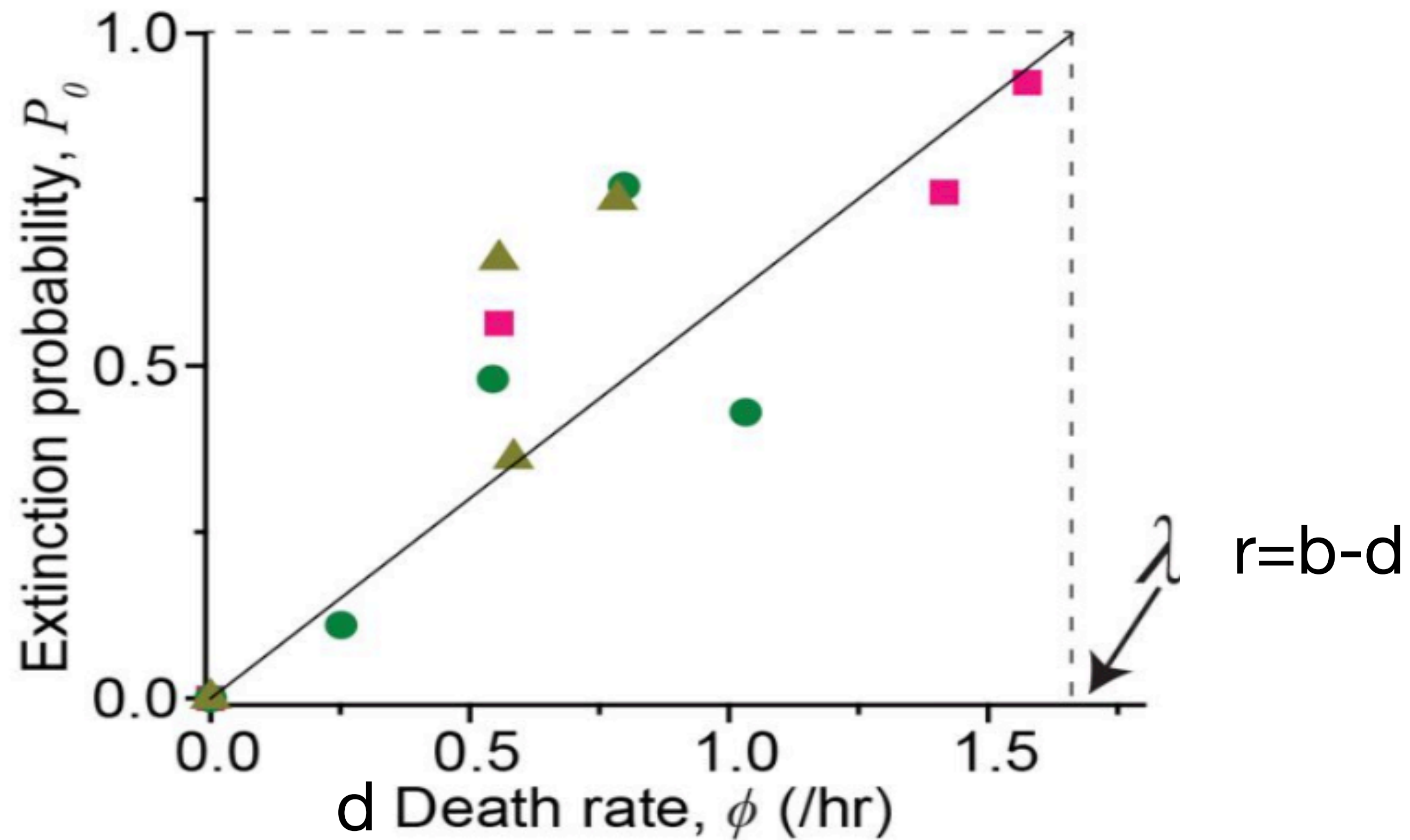
Estimating division and death rate



Death rate is modulated, growth rate ~not



Extinction probability is d/b



killing takes time and
and is stochastic

